

The Proof

She had been working on the proof for nine years. She was three weeks from publication when she found the gap.

CEFR Level B2

Book 9 of 10 • Mathematical Drama / Intellectual Thriller • Adult Graded Reader Series

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About This Book

Setting: A university mathematics department. Nine years compressed into three weeks.

Genre: Mathematical Drama / Intellectual Thriller

Characters

- Dr. Sofia — 44. Professor of pure mathematics. The proof concerns a conjecture in algebraic topology.
- Nathan — 36. Her postdoc. The one who finds the gap. Methodical, scrupulously honest.
- The conjecture — forty years old, attributed to a mathematician who died without proving it.
- The gap — a flaw in the argument at page seventy-three. Small. Possibly fatal.

THE PROOF

Page 1 — Three Weeks

The paper was three weeks from submission when Nathan knocked on her door.

She had been expecting congratulations — the department knew the paper was nearly ready, and the anticipation that had accumulated over nine years of work had, in the final month, produced a quality of collective suspended breath that she had found simultaneously gratifying and unsettling.

Nathan did not look like someone who had come to congratulate her.

He looked like someone who had been awake for most of the previous thirty-six hours and had, in the course of that sleeplessness, found something he had not been looking for and did not know how to give back.

'I need to show you something,' he said.

She had known, from the tone, before she saw it.

Vocabulary Support

submission	gönderim
conjecture	konjektür, varsayım
gratifying	tatmin edici

suspended breath	kesilen nefes
sleeplessness	uykusuzluk
scrupulously honest	titizlikle dürüst

She knew from the tone, before she saw it. Nine years. Three weeks. A knock.

Page 2 — Page Seventy-Three

The gap was on page seventy-three of the ninety-four-page paper.

It was, Nathan had taken three pages of his own notation to describe, a flaw in the argument at the juncture between the third and fourth sections — a place where she had made an assumption that was necessary for the argument to proceed and which was, he believed, not adequately justified.

She read his description of the gap three times.

She then read the relevant section of the paper, which she knew as well as she knew any piece of mathematics she had ever produced.

She found, on the third reading, that he was right.

The gap was real.

Whether it was fatal was a different question.

Vocabulary Support

notation	notasyon, gösterim
juncture	birleşim noktası
adequately justified	yeterince gerekçelendirilmiş
relevant section	ilgili bölüm
proceed	devam etmek
flaw	kusur, hata

He was right. The gap was real. Whether it was fatal was a different question.

Page 3 — What the Gap Was

She had been working in algebraic topology long enough to know that not all gaps were equal.

Some gaps were genuinely fatal — places where the argument broke down in ways that could not be repaired without restructuring the proof from a much earlier point, sometimes invalidating years of work.

Some gaps were remediable — places where an assumption had been made implicitly that could be made explicit, where an argument had been compressed in ways that could be expanded, where the gap was a presentational problem rather than a mathematical one.

And some gaps were apparent rather than real — places where the argument was in fact correct but had been expressed in a way that made the correctness non-obvious.

Nathan's gap was, she could not yet determine, one of the three.

Vocabulary Support

algebraic topology	cebirsel topoloji
remediable	giderilebilir
invalidating	geçersizleştirmek
implicit	zımni, örtük

restructuring	yeniden yapılandırma
apparent	görünürde

Fatal, remediable, or apparent. She could not yet determine which. That was the three weeks.

Page 4 — Nathan

She had taken Nathan on as a postdoc eighteen months ago, specifically for his reputation for finding problems that other people had missed.

It was not a reputation that made him popular among colleagues who wanted their work confirmed rather than tested. It made him, in her view, invaluable.

He had spent three months going through the paper — not as a favour, but because she had asked him to, because she had learned, over nine years, that work done in isolation acquired blind spots that only a fresh and adversarial eye could identify.

'Did you find anything else?' she asked, after the silence that followed his presentation of the gap.

'No,' he said. 'This is the only one.'

'Are you sure?' 'I'm as sure as I can be. That doesn't mean I'm certain.'

'No,' she said. 'It doesn't.'

Vocabulary Support

postdoc	doktora sonrası araştırmacı
adversarial	karşı çıkıcı
invaluable	paha biçilmez
blind spots	kör noktalar

isolation	yalnızlık, izolasyon
confirmed	doğrulanmış

"As sure as I can be. Not certain." The right epistemic humility. She had hired him for this.

Page 5 — The First Week

She spent the first week attempting to repair the gap.

The approach she took was the standard one: assume the gap was remediable, and attempt to construct the missing justification.

She worked for twelve hours a day, which was more than she had worked in years — not because of urgency, though urgency was present, but because the problem had a quality that mathematical problems sometimes had: it pulled.

By the end of the first week, she had produced three partial approaches, none of which had resolved the gap and all of which had, in the process of their failure, clarified what the gap actually was.

This was progress.

It was not the progress she had hoped for.

Vocabulary Support

remediable	giderilebilir
partial approaches	kısmi yaklaşımlar
clarified	açıklık kazandırdı
urgency	aciliyet
pulls (a problem)	çeker (bir problem)
construct the justification	gerekçeyi inşa etmek

Three partial failures, all of which clarified what the gap actually was. Progress. Not the hoped-for kind.

Page 6 — The Department

She had not told the department about the gap.

This was not deception — nothing had been announced, no submission had been made, no one had a claim on a timeline that had not been formally established.

But the accumulated anticipation she had been aware of for months was now, from her perspective, something she was managing rather than inhabiting.

Her colleagues asked how the paper was going.

She said: 'Well. Still working.'

This was true. It was also, in its compression, a form of concealment that she found distasteful but necessary.

She thought: mathematics does not allow social management of its problems.

The problem either is or is not. The proof is or is not. No amount of not mentioning it changes that.

Vocabulary Support

deception	aldatma
accumulated anticipation	birikmiş beklenti
concealment	gizleme
distasteful	rahatsız edici
inhabiting	içinde yaşamak

social management	sosyal yönetim
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*Mathematics does not allow social management of its problems.
She knows this. She manages anyway.*

Page 7 — What She Remembered

In the middle of the second week, she remembered something.

She remembered a seminar she had attended eleven years ago — not the main content of the seminar, but a remark made by the speaker during questions, which she had written in her notebook and which had not, at the time, seemed significant.

The notebook was in the fourth drawer of her filing cabinet.

The remark, when she found it, read: 'The difficulty with this class of problem is that the gap and the proof of the gap are often structurally related — the thing that makes the gap appear also contains the structure needed to close it.'

She read it twice.

She thought: the speaker was right. And I may have been working in the wrong direction.

Vocabulary Support

seminar	seminer
structurally related	yapısal olarak ilişkili
filing cabinet	dosya dolabı
close a gap	açığı kapatmak
wrong direction	yanlış yön

class of problem	problem sınıfı
<i>The gap and its proof are structurally related. She had been working in the wrong direction.</i>	

Page 8 — The Approach

The new approach required her to reframe the gap not as a deficiency to be remedied from outside but as a structural feature of the argument whose properties she had not yet fully characterised.

This was, she understood as she began working on it, a fundamentally different kind of problem from the one she had been trying to solve.

Instead of building a bridge across the gap, she was attempting to understand what the gap was made of.

Nathan had joined her on the second day of the new approach. He had not needed to be asked.

'This is different,' he said, looking at the new notation.

'Yes,' she said. 'I think I've been treating it as an obstacle. It might be a feature.'

'A feature of what?' 'Of the argument itself. I'm not sure yet.'

Vocabulary Support

deficiency	eksiklik
reframe	yeniden çerçevelemek
characterised	karakterize edilmiş
fundamentally different	temelden farklı
obstacle	engel

feature	özellik
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"Not an obstacle — a feature." The gap might be part of the argument, not a failure of it.

Page 9 — Day Fourteen

On the fourteenth day — the last day of the second week — something happened that she had experienced twice before in her mathematical life and which she had, each time, found impossible to describe accurately.

She was not, at that point, solving the problem. She was working through a subsidiary calculation that she expected to inform the main argument but not to resolve it.

The subsidiary calculation produced an identity.

The identity was not what she had expected.

She looked at it for a long time.

She thought: if this is correct, and if the conclusion I think it implies is correct, then the gap is not a gap. It is a theorem.

She checked the calculation four times before she allowed herself to say this to Nathan.

Vocabulary Support

subsidiary calculation	ikincil hesaplama
identity	özdeşlik
implies	ima eder
theorem	teorem

informed the main argument	ana argümanı bilgilendirmek
accurately	doğru biçimde
<i>The gap is not a gap. It is a theorem. She checked four times before saying it aloud.</i>	

Page 10 — What It Meant

What it meant, if she was right, was this.

The gap was not an error in the argument. It was an unexplored consequence of the argument — a result that the main theorem implied but that she had not recognised as a theorem in its own right because she had been, for nine years, attending to the main result rather than to the structures she was building to reach it.

The gap was, in other words, a discovery hidden inside the proof of a larger discovery.

Nathan had spent the rest of the day checking her work.

He had, at seven in the evening, looked up from the pages and said: 'I think you're right.' A pause. 'I think you might have found two theorems.'

She had looked at the window for a moment.

'Nine years,' she said. Not as a complaint.

Vocabulary Support

consequence	sonuç
unexplored	keşfedilmemiş
in its own right	kendi başına
attending to	... e dikkat etmek
hidden inside	içinde gizli

structures	yapılar
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"Nine years." Not as a complaint. As something else she did not have a word for.

Page 11 — The Rewriting

The paper required significant restructuring.

The gap, reframed as a theorem, needed to be proved independently and its relationship to the main result made explicit.

This added, she estimated, approximately fifteen pages and three weeks of work.

She had, by this point, entirely lost track of the original three-week timeline.

The department's collective anticipation, which had previously felt like pressure, had receded into the background.

She was working on a problem again, rather than managing the completion of a project.

There was, she noticed, a qualitative difference in how this felt.

She had not felt it in some time.

Vocabulary Support

qualitative difference	niteliksel fark
receded into the background	arka plana çekildi
independently	bağımsız olarak
explicit	açık, net
estimated	tahmin etti

entirely lost track

tamamen takibini kaybetti

A qualitative difference in how it felt. Working on a problem again, not managing a completion.

Page 12 — What Nathan Said

Nathan had, during the three weeks, become a collaborator in a way that the postdoc relationship did not formally require.

She had not asked him to work the hours he had worked, and she had been aware, as the weeks progressed, that his investment in the problem was no longer the investment of someone discharging a professional obligation but something more.

She had asked him, on the seventeenth day, why he was still working so late.

He had said: 'Because it's interesting. And because the problem has good structure.' A pause. 'And because I want to see how it comes out.'

She had looked at him.

'So do I,' she said. 'After nine years, I still don't know how it comes out.'

He had smiled. 'That,' he said, 'is what pure mathematics is.'

Vocabulary Support

discharging an obligation	bir yükümlülüğü yerine getirmek
collaborator	iş birlikçi
investment in	... e yatırım
good structure	iyi yapı

pure mathematics	saf matematik
formally required	resmi olarak zorunlu

"After nine years, I still don't know how it comes out." That is what pure mathematics is.

Page 13 — The Question of Credit

The question of authorship had not been raised.

She raised it herself, in the fourth week, because it required raising and because she did not want it to be raised by Nathan or by the institution.

She said: 'The theorem that we — that came from the gap — is partly yours. You found the gap. Your reading of the structure led to the reframing. I think you should be a co-author on the paper.'

Nathan was quiet for a moment.

'The gap-finding was not mathematical contribution,' he said. 'It was checking. That's what I was hired for.'

'The reframing was,' she said. 'The insight that it was structural rather than deficient was a contribution.'

'That was yours,' he said.

'It followed from your finding,' she said. 'We can discuss proportionality. We cannot not discuss it.'

Vocabulary Support

authorship	yazarlık
co-author	ortak yazar
mathematical contribution	matematiksel katkı

reframing	yeniden çerçeveleme
deficient	eksikli, yetersiz
proportionality	orantısallık

"We cannot not discuss it." She raises what others avoid. She would not be a good mathematician if she did not.

Page 14 — The Resolution

The resolution, reached over two conversations, was that Nathan would be listed as second author.

He had argued for an acknowledgement rather than authorship, and she had argued back, and they had arrived — through a process she found more intellectually rigorous than most of the professional negotiations she had conducted — at a position they could both justify.

The justification was not, in the end, about credit or fairness in the abstract. It was about accuracy.

'The paper is more accurately attributed,' she said, 'with your name on it. It represents what actually happened.'

Nathan had looked at her for a long time.

'You're the only mathematician I know,' he said, 'who applies the same standards to authorship that they apply to the mathematics.'

'They're the same problem,' she said.

Vocabulary Support

acknowledgement	teşekkür bölümü
rigorous	titiz
attributed	atfedilen
fairness in the abstract	soyut adalet

justification	gerekçe
listed as	... olarak listelenmek

"They're the same problem." Authorship and mathematics. The same standard of accuracy.

Page 15 — The Submission

The paper was submitted six weeks after Nathan's knock on the door.

It was ninety-seven pages, credited to two authors.

The abstract contained, in its final sentence, a statement she had not expected to be writing when the process began: 'A subsidiary result, arising from the investigation of a structural gap in the main argument, establishes a previously unproved consequence of the conjecture.'

She read the abstract one final time before submission.

She thought: this is the paper that the nine years made.

She thought: not the paper I set out to write.

She thought: better.

She pressed submit.

Vocabulary Support

abstract	özet
subsidiary result	ikincil sonuç
consequence	sonuç
set out to write	yazmaya niyetlenmek
previously unproved	daha önce kanıtlanmamış
establishes	ortaya koyar

"Better." Not the paper she set out to write. Better. She pressed submit.

Page 16 — The Review

The paper was under review for four months.

She had submitted to a journal with the longest and most rigorous review process in the field — a journal where papers were reviewed by three referees, any one of whom could require revisions.

The reviews came back with comments that required three weeks of additional work.

Two of the reviewers had found the new theorem interesting and had asked for more careful exposition of its relationship to the main result.

One reviewer had found what they described as a potential gap in the proof of the subsidiary theorem.

Nathan had looked at the reviewer's comment.

'It's not a gap,' he said. 'It's the same structural phenomenon. We need to prove it there too.'

She looked at him. 'Yes,' she said. 'We do.'

Vocabulary Support

referees	hakemler
exposition	açıklama, sunum
structural phenomenon	yapısal fenomen
revisions	revizyonlar
rigorous review	titiz değerlendirme
potential gap	potansiyel açık

*The same structural phenomenon. They need to prove it there too.
The gap teaches them to look for it elsewhere.*

Page 17 — Publication

The paper was published eight months after the original three-week deadline.

The response in the field was, she had been told to expect, a period of careful reading followed by either silence, questions, or confirmation.

What happened was questions.

Several colleagues wrote to her within the first month — not with objections, but with observations that the subsidiary theorem had implications she had not fully worked out.

She found this gratifying in a way that the original paper's expected acclaim had not quite been.

She had expected acclaim for completing something.

The questions meant that she had begun something.

They were not, she thought, the same thing.

Vocabulary Support

confirmation	doğrulama
implications	çıkarımlar
objections	itirazlar
worked out	çözülmüş
gratifying	tatmin edici
period of careful reading	dikkatli okuma dönemi

Completing something vs beginning something. The questions meant she had begun. Better.

Page 18 — What Nathan Did

Nathan had, in the months after publication, begun a paper of his own.

It concerned, she learned when he described it, a generalisation of the subsidiary theorem — a broader result of which their theorem was a special case.

She had read the draft. It was good.

She had said so, and had added: 'You should publish this independently. This is your work.'

He had said: 'It came from your paper.' 'Yes,' she said. 'Most important things come from somewhere. That doesn't change whose they are now.'

He had looked at her.

'I wouldn't have found it,' he said, 'without the gap.'

'Neither would I,' she said. 'That's what I mean.'

Vocabulary Support

generalisation	genelleme
special case	özel durum
independently	bağımsız olarak
came from	... den geldi
broader result	daha geniş sonuç
whose they are now	şimdi kimin oldukları

*"That's what I mean." The gap generated both papers. Both real.
The gap was the gift.*

Page 19 — The Gap

She had been asked, at a conference, to speak about the process of the proof.

She had given a technical account of the mathematical content, which was what most people wanted.

At the end, she had added something she had not planned to add.

She said: 'The most significant part of this work was not the main theorem. It was a gap found three weeks before submission — a place where the argument was insufficiently justified. The attempt to close the gap produced a second theorem. I have been asked whether this was a fortunate accident. I think the question misframes what happened. Mathematics is not a process of building correct arguments from scratch. It is a process of finding out where your understanding is incomplete and using the incompleteness to locate what you did not know you were looking for.'

The room was quiet.

She thought: nine years. One gap. Two theorems.

She thought: that is mathematics.

Vocabulary Support

insufficiently justified	yetersiz gerekçelendirilmiş
fortunate accident	şanslı kaza
misframes	yanlış çerçeveler
incompleteness	eksiklik
locate	bulmak
from scratch	sıfırdan

"Using the incompleteness to locate what you did not know you were looking for." That is mathematics.

Page 20 — The Conjecture

The conjecture had been posed forty years ago by a mathematician who had spent twenty years working on it and had died, in 1994, without a proof.

Sofia had been nineteen when he died. She had not, at that point, begun to work in the relevant area.

She had found his notebooks in a university archive six years into her own work — notebooks full of approaches he had abandoned, false starts, observations that he had not been able to connect to the central problem.

In one of them, he had written, in a margin: 'The difficulty is probably structural. I suspect the answer is somewhere in the structure of the gap itself.'

She had read this the first time as an expression of frustration.

She had read it again, after the proof was published, as something else.

He had known. He had just not known what he knew.

Vocabulary Support

posed	ortaya atılmış
abandoned	terk edilmiş
false starts	sahte başlangıçlar

in a margin	kenarda, kenara
expression of frustration	hayal kırıklığı ifadesi
structural	yapısal

"He had known. He had just not known what he knew." Forty years. The gap was always the answer.

ACTIVITIES

1. Vocabulary Practice — Match the Words

Match each word or phrase with the correct meaning.

Word / phrase	Meaning
1. conjecture	A. konjektür
2. theorem	B. teorem
3. remediable	C. giderilebilir
4. subsidiary	D. ikincil
5. adversarial	E. karşı çıkıcı
6. generalisation	F. genelleme
7. exposition	G. açıklama
8. incompleteness	H. eksiklik

2. Fill in the Blanks

Choose the correct word or phrase from the box.

gap — feature — two — structure — began — accuracy

1. Nathan found a _____ in the argument at page seventy-three.
2. Sofia reframed the gap as a _____ of the argument rather than a deficiency.
3. The gap was not a gap — it was a theorem. The paper ended up containing _____ theorems.

4. The speaker at the seminar eleven years ago had said: the gap and its proof are _____ally related.
5. She said authorship and mathematics were the same problem — both required _____.
6. The questions from colleagues meant she had _____ something, not completed something.

3. Comprehension Questions

True or False? Write T or F.

1. Nathan finds multiple gaps in the proof. ____
2. The gap is on the last page of the paper. ____
3. Nathan suggests reframing the gap as a structural feature. ____
4. Sofia decides to name the new theorem after Nathan. ____
5. The paper is published with Nathan as second author. ____
6. Nathan begins a new paper based on a generalisation of the subsidiary theorem. ____

Short Answer Questions

7. What are the three types of gap Sofia identifies, and which does she suspect Nathan's gap is?
8. What does the remark from the eleven-year-old seminar contribute to Sofia's approach?
9. What does Sofia mean when she says the gap turned out to be a 'feature' of the argument?
10. Why does Sofia insist on listing Nathan as co-author?
11. How does Nathan describe his reason for working late during the three weeks?
12. How does Sofia describe mathematics at the conference?
13. What does Sofia find in the archived notebooks of the mathematician who posed the conjecture?

14. What does Sofia mean when she says 'completing something' and 'beginning something' are not the same?

15. What does the last line mean — 'he had known. He had just not known what he knew'?

ANSWER KEY

(Alıştırmaları yaptıktan sonra bakınız.)

Match: 1-A 2-B 3-C 4-D 5-E 6-F 7-G 8-H

Fill in the Blanks: 1. gap 2. feature 3. two 4. structure 5. accuracy 6. began

True / False: 1. F 2. F 3. F 4. F 5. T 6. T

7. Fatal (breaks the argument irreparably), remediable (missing justification can be added), or apparent (correct but non-obviously expressed). She cannot yet determine which Nathan's gap is.

8. It suggests that the gap and its proof are structurally related — the thing making the gap appear may contain the structure needed to close it. This leads her to reframe the gap as a structural feature rather than a deficiency.

9. The gap is not a missing step that needs to be bridged from outside the argument — it is an unexplored consequence of the argument itself, a result the main theorem implies but that she had not recognised as a theorem in its own right.

10. Because accuracy requires it — the paper represents what actually happened, and Nathan's contribution to the reframing was a genuine mathematical insight. Authorship and mathematics are the same problem: both require accuracy.

11. Because the problem has good structure, and because he wants to see how it comes out. Pure mathematics.

12. Mathematics is not building correct arguments from scratch — it is finding where your understanding is incomplete and using the incompleteness to locate what you did not know you were looking for.

13. He had written in a margin that the difficulty was probably structural and that the answer was somewhere in the structure of the gap itself — which was precisely the insight that produced the second theorem.

14. Completing something means bringing an established project to closure. Beginning something means opening new questions — the questions from colleagues meant the paper had generated new problems to investigate.

15. The original mathematician had correctly intuited that the gap was structural and contained the answer — but he had not had the mathematical tools or the years of work needed to articulate or prove what he was intuiting. He knew in the intuitive sense but not in the mathematical sense.